Jiheng Liang

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Education

Ph.D. Student in Data Science — 2025 – 2029 (Expected)
William & Mary, Williamsburg, Virginia, USA

M.Sc. in Physics — 2022 – 2025
Sun Yat-sen University, Guangzhou, China

B.Sc. in Optoelectronics Information Science and Engineering — 2018 – 2022
Sun Yat-sen University, Guangzhou, China

Research Interests

Graph-to-Text, Graph Representation Learning, Large Language Models for Graphs, LLM Evaluation, Bioinformatics, Spectroscopy

Research Experience

Research Assistant — 2025 – Present
William & Mary, Williamsburg, Virginia, USA

Publications

Manuscripts

ICLR'27 submitted to the International Conference on Learning Representations, 2027. Under review. First author.

WWW'27 submitted to The ACM Web Conference, 2027. Under review. First author.

WWW'27 submitted to The ACM Web Conference, 2027. Under review.

arXiv **Jiheng Liang**, Zujie Xie, Ziru Yu, and Xiangyang Yu. "Enhancing Spectral Knowledge Interrogation: A Reliable Retrieval-Augmented Generative Framework on Large Language Models." arXiv:2408.11557, 2024; revised 2025. [Paper]

Peer-Reviewed Journal Articles

ACA Ziru Yu*, **Jiheng Liang***, Zujie Xie, and Jing He. "SpectralTAG: Functional group identification from infrared spectra via unified LLM and GNN textual graphs." Analytica Chimica Acta, vol. 1424, article 346160, 2026 (forthcoming issue). DOI: 10.1016/j.aca.2026.346160 *Co-first authors.

ACA Zujie Xie, Zixuan Chen, **Jiheng Liang**, Xiangyang Yu, and Ziru Yu. "LUMIR: an LLM-driven unified agent framework for multi-task infrared spectroscopy reasoning." Analytica Chimica Acta, vol. 1382, article 344857, 2026. DOI: 10.1016/j.aca.2025.344857

SAA **Jiheng Liang**, Xiangyang Yu, Weibin Hong, and Yefan Cai. "Information extraction of UV-NIR spectral data in waste water based on Large Language Model." Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy, vol. 318, article 124475, 2024. DOI: 10.1016/j.saa.2024.124475

Conference Papers

NeurIPS'26 **Jiheng Liang**, Chen Zhao, Di Wu, Chenyang Bu, Yunpeng Hong, Xingquan Zhu, and Yi He. "ColdDDI: Evaluating Knowledge Utilization in Cold-Start Drug-Drug Interaction Prediction." Conference on Neural Information Processing Systems, Evaluations & Datasets Track, 2026. Accepted, poster. [Paper] [Code]

BIBM'26 **Jiheng Liang** and Chen Zhao. "DDI-LoRA: A Low-Rank Semantic Adapter for Cold-Start Drug–Drug Interaction Prediction." IEEE International Conference on Bioinformatics and Biomedicine, Doctoral Forum, 2026. Accepted.

IFAL'24 **Jiheng Liang**, Zujie Xie, Ziru Yu, Yefan Cai, Longzhao Lu, and Xiangyang Yu. "Near-field hyperspectral technology and its application in automotive lighting." 12th International Forum of Automotive Lighting, Shanghai, China, October 16–18, 2024.

Professional Service

Journal Referee

- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy (SAA)

Conference Reviewer

- IEEE International Conference on Bioinformatics and Biomedicine (BIBM)
- The ACM Web Conference (WWW) 2027